

Assignment 1 – Part 2 | Decision Trees

This report provides an overview of the Decision Tree model used to classify real vs. fake news headlines. First the `load_data` function stores the contents of the fake and real text files in variables that are then given labels; 1 and 0 for real and fake, respectively. We then convert the text to numerical features using the `CountVectorizer()` method from `sklearn.feature_extraction.text` module and split the dataset randomly into 70% training, 15% validation, and 15% test examples.

1 Results and Visualizations

The training, validation, and test X and Y variables are then passed into the `select_model()` function which trains the decision tree classifier using the following values of `max_depth` -> [2,4,6,8,10], evaluating the performance of each one on the validation set resulting in the following results:

Table 1. Resulting validation accuracy for each maximum depth value

Max Depth	Validation Accuracy
2	0.6857
4	0.6796
6	0.6816
8	0.6857
10	0.6980

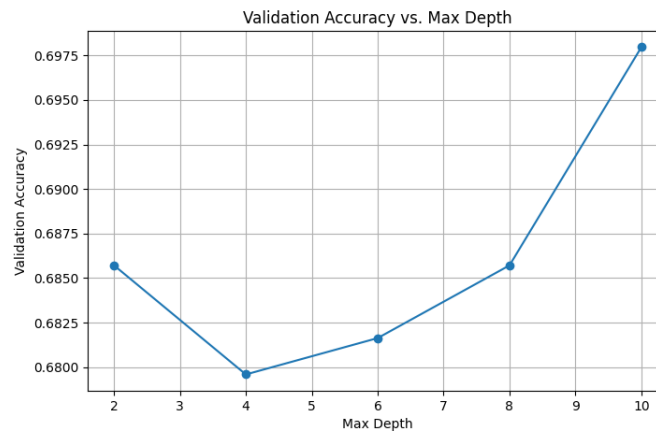


Figure 1. Max depth plotted against validation accuracy

After analyzing the results, we observe the depth with the best accuracy is a depth of **10**, with an accuracy of **0.7184**. Below we visualize the first two layers of the tree with the hyperparameters which achieved the highest validation accuracy.

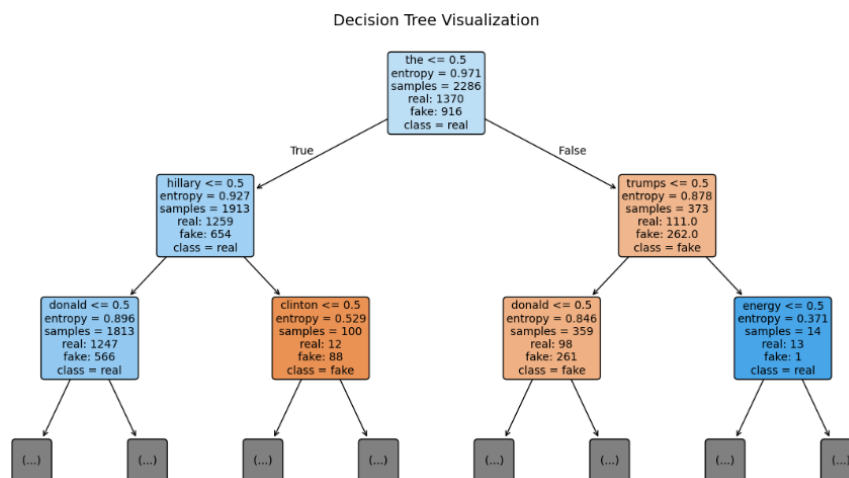


Figure 2. Visualization of the first two layers of the tree. We observe how each decision node corresponds to a word feature, splitting on the existence or absence of words like “Hillary”, “Clinton”, “Donald”, “Trump”. Also shows the sample size at each node and the split between real and fake, identifying a class, based on the split.